

DSA 8420 Spring 2025

Homework 5

Due February 18, 2025

1. Consider the following linear program.

$$\begin{array}{ll}\max & z = -3x_1 + 4x_2 + x_3 + 2x_4 + 3x_5 \\s.t. & x_1 + x_2 + x_3 - x_4 + 2x_5 \leq 12 \\& x_1 - 2x_2 - x_3 - x_4 - x_5 \geq -30 \\& x_1, x_2, x_3, x_4, x_5 \geq 0\end{array}$$

- (a) (4 points) Transform the problem into standard form.
(b) (12 points) Use the simplex method to solve the problem.

2. Consider the following linear program.

$$\begin{array}{ll}\max & z = 2x_1 + 2x_2 \\s.t. & x_2 \leq 10 \\& x_1 - 3x_2 \leq 2 \\& x_1 + x_2 \leq 16 \\& x_1, x_2 \geq 0\end{array}$$

- (a) (4 points) Transform the problem into standard form.
(b) (14 points) Use the simplex method to find two optimal basic feasible solutions.

3. Consider the following linear program.

$$\begin{array}{ll}\max z = & 4x_1 + x_2 \\s.t. & 2x_1 - 3x_2 \leq 12 \\& -4x_1 + x_2 \leq 8 \\& x_1, x_2 \geq 0\end{array}$$

- (a) (4 points) Transform the problem into standard form.
- (b) (12 points) Use the simplex method to show that the linear program in (a) is unbounded.